Near-Eye Purkinje Image Simulation and Capture for Accommodation-Aware Eye Tracking

Abstract

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**;
• **Computing methodologies** → **Rendering**; *Computer vision.*

Keywords

eye tracking, deep learning, computer graphics

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1 Problem

1.1 Challenges of reconstructing Purkinje reflections via 3D Gaussian Splatting (3DGS).

3DGS represents view-dependent appearance by using low-order Spherical Harmonics (SH) [Kerbl et al. 2023a], therefore, accurately modeling high-frequency, view-dependent specular appearance remains challenging [Jiang et al. 2024a; Tang et al. 2025]. More importantly, P3 and P4 are formed through transmission and reflection across the multiple ocular interfaces of eyes, whereas rasterization-based 3DGS does not explicitly model these secondary optical paths [Wu et al. 2025] (see Fig. X). In addition, Lu et al. [Lu et al. 2020] and Wu et al. [Cornsweet and Crane 1973] show that P3 is formed at a different optical focal plane from P1 and P4, while higher-order Purkinje reflections are relatively weak and therefore more susceptible to defocus and low contrast. Finally, Purkinje locations vary jointly with accommodation, vergence, subject-specific ocular geometry, and camera–illumination configuration [Lu et al. 2020; Ozhan et al. 2023]. Purkinje reflections are better characterized as dynamic optical signals governed by ocular optics rather than stable surface textures, making it difficult for standard 3DGS to reconstruct them consistently with their underlying physical behavior.

2 Introduction

3 Related works

To compare our work with existing eye convergence and accommodation tracking, we review related works from two main areas: physical feature-based experimental designs and virtual eye modeling and neural rendering. To clearly show the technical features of previous works, Table 1 categorizes and compares these methods based on their hardware setups, algorithms, operating modes, and tracking accuracy.

3.1 Eye Accommodation Tracking

In stereoscopic and near-eye displays, the Vergence-Accommodation Conflict (VAC) [Hoffman et al. 2008]arises when the eyes converge to a virtual 3D depth while the crystalline lens remains focused on the physical display plane. This mismatch can reduce fusion performance and cause visual discomfort . Recent VAC-related depth-perception studies therefore use eye-tracking and controlled optical setups to measure vergence responses under different perceived-depth conditions, highlighting the need for objective and reproducible eye-based measurements in VAC-related experiments [Bontha and Arefin 2024; Sturgeon et al. 2025]. However, vergence alone does not fully describe the accommodative state of the eye. Accommodation is harder to measure directly, especially in compact near-eye systems. Higher-order Purkinje images provide a promising optical cue because they arise from reflections at ocular interfaces, including the crystalline lens surfaces, and vary with the lens geometry during accommodation. Lu et al.[2020] use a multi-camera, multi-LED AR-glasses prototype to track higher-order Purkinje images, extending conventional pupil–corneal-reflection tracking toward accommodation-aware sensing in a near-eye form factor. Ozhan et al.[2023] extend this direction by collecting multi-subject Purkinje measurements and combining a Purkinje imaging setup with a **Multi-Layer Perceptron (MLP)**-based regression model to estimate dynamic vergence and accommodation across users.

These works suggest that higher-order Purkinje reflections can serve as physically meaningful cues for accommodation-aware eye tracking, but they also reveal a practical challenge: P3 and P4 are faint, easily defocused, and highly sensitive to LED–camera–eye geometry. Motivated by this limitation, we first build a controllable 3D optical eye model to analyze the formation, visibility, and stability of P1–P4 under near-eye infrared imaging conditions.

3.2 Computational Eye Modeling

Computational eye models are widely used to generate labeled data for gaze estimation. UnityEyes, NVGaze, and RIT-Eyes synthesize controllable variations in gaze, ocular geometry, appearance, and illumination for training eye-tracking models [Kim et al. 2019b; Nair et al. 2020; Wood et al. 2016]. These systems capture visible eye-region cues effectively, but they primarily target eye rotation, pupil localization, and segmentation rather than accommodation-dependent changes within the crystalline lens.

Neural eye representations improve visual realism and relighting. EyeNeRF combines an explicit eyeball surface with implicit volumetric representations for view synthesis, gaze animation, corneal refraction, and relighting [Li et al. 2022]. Khademi et al. extend a similar hybrid formulation to near-field point sources and structured illumination [Khademi et al. 2026]. However, these methods focus

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Table 1: Comparison of representative approaches related to vergence (V) and accommodation (A) sensing. Colours rate each entry *relative to the other options in its column* with respect to one goal, a compact wearable sensor that estimates V and A jointly; green is better suited, yellow intermediate, red least suited. Camera: imaging hardware per eye, ordered by the minimum needed for binocular V and A – one camera per eye (the lower bound, since V requires both eyes and A one) > two or more sensors per eye > bench-top optics with head stabilisation. This ordering reflects footprint, not accuracy: head-fixed bench systems and multi-view rigs achieve lower errors by removing or measuring eye–camera motion, whereas a single camera must rely on illumination layout and eye-model priors (see Accuracy). Illum.: integrability and designability of the source geometry – freely positionable IR/NIR LEDs, whose number and placement are chosen for the sensing principle, > LEDs fixed inside a commercial tracker, placed by the vendor for corneal glints and not adjustable, > bench-top collimated or off-axis sources tied to a working distance. As for Camera, this reflects integrability, not image quality: collimated bench sources yield the sharpest lens reflections. Method: whether the signal physically depends on lens state and can be captured by a compact camera – Purkinje reflections (lens-dependent, camera and LED only) > photorefraction (lens-dependent, but requires an eccentric source at a bench working distance) > pupil/glint gaze tracking (no A information). Data: validated on real eyes > synthetic only. Learning: handling of inter-subject variability – learned models > physical eye models requiring per-user calibration > fixed formulas such as binocular triangulation. V/A Target: quantity directly estimated (both > one > neither). Near-eye: demonstrated in a physical near-eye configuration. Accuracy: error reported for the quantity in parentheses – gaze: high $< 0.5°$, mid $0.5–3°$, low $> 3°$; A and V (depth in dioptres): high < 0.25 D, mid $0.25–0.5$ D, low > 0.5 D; protocols differ across works, so values are indicative only.

Ref.	Camera	Illum.	Method	Data	Learning	V/A Target	Near-eye	Accuracy
[Cornsweet and Crane 1973]	Bench	Bench source	Purkinje	Real	Physical model	Neither	No	High (gaze)
[Crane and Steele 1978]	Bench	Bench source	Purkinje	Real	Physical model	A only	No	High (A)
[Schaeffel et al. 1987]	Bench	Bench source	Photorefraction	Real	Physical model	Both	No	Mid (A)
[Morimoto et al. 2002]	1/eye	Movable LEDs	Pupil/glint	Real	Physical model	Neither	No	Mid (gaze)
[Rosales et al. 2006]	Bench	Bench source	Purkinje	Real	Physical model	A only	No	Mid (A)
[Świrski and Dodgson 2013]	1/eye	Fixed LEDs	Pupil/glint	Real	Physical model	Neither	Yes	Mid (gaze)
[Weier et al. 2018]	1/eye	Fixed LEDs	Pupil/glint	Real	Regression	V only	Yes	Mid (V)
[Kim et al. 2019b]	1/eye	Movable LEDs	Pupil/glint	Both	CNN	Neither	Yes	Mid (gaze)
[Padmanaban et al. 2019]	1/eye	Fixed LEDs	Pupil/glint	Real	Formula	V only	Yes	Mid (V)
[Lu et al. 2020]	≥2/eye	Movable LEDs	Purkinje	Synthetic	NN regression	Both	Yes	Mid (A)
[Angelopoulos et al. 2021]	≥2/eye	Movable LEDs	Pupil/glint	Real	Physical model	Neither	Yes	High (gaze)
[Wu et al. 2023]	Bench	Bench source	Purkinje	Real	Physical model	Neither	No	High (gaze)
[Ozhan et al. 2023]	1/eye	Movable LEDs	Purkinje	Real	ML regression	Both	No	High (A)
[Ahn and Lee 2025]	1/eye	Movable LEDs	Purkinje	Real	Regression	V only	No	Low (V)
[Sturgeon et al. 2025]	1/eye	Fixed LEDs	Pupil/glint	Real	Formula	V only	Yes	Low (V)
[Ressmeyer et al. 2026]	Bench	Bench source	Purkinje	Real	Physical model	Neither	No	High (gaze)
Ours	1/eye	Movable LEDs	Purkinje	Both	3DGS	Both	Yes	High (A)

on photorealistic appearance and do not explicitly predict P3/P4 formation or accommodation-dependent internal ocular geometry.

Purkinje-based systems model internal reflections more directly. Wu et al., Idrizovi'c et al., and OpenIrisDPI use P1 and P4 for accurate eye-rotation tracking [Idrizović et al. 2025; Ressmeyer et al. 2026; Wu et al. 2023]. Lu et al. further model higher-order reflections with multiple cameras and infrared LEDs to estimate vergence and accommodation [Lu et al. 2020]. Ozhan et al. simulate P1–P4 across accommodation and vergence states, validate the predictions against real captures, and use the resulting features for dynamic estimation [Ozhan et al. 2023]. Our work follows this direction by using a controllable optical eye model to predict how accommodation, vergence, and LED–camera geometry affect P1–P4 formation.

3.3 Physics-Based Optical Design

Differentiable optical design provides a possible framework for optimizing the illumination and imaging layout of a Purkinje capture system. Na et al. show that image-domain losses can be propagated through differentiable ray tracing to physical optical parameters [Na et al. 2024]. For our system, the same principle could be applied to LED position, camera pose, focal distance, aperture, and field of view.

The optimization criteria should follow the capture behavior of higher-order Purkinje reflections. Lu et al. show that P3 is weaker than P1, lies on a different focal plane, and may leave the camera field of view as gaze changes; their system therefore uses multiple cameras and infrared LEDs [Lu et al. 2020]. Ozhan et al. further show that the camera–LED–eye geometry affects the observed P3/P4 locations and configure the system to improve reflection separation while avoiding vignetting [Ozhan et al. 2023]. These results suggest optimizing P3/P4 visibility, sharpness, field-of-view coverage, and separation from other reflections.

The selected layout must also remain practical to assemble, calibrate, and evaluate at the sensor level. Prior work on replicating a controllable AR haploscope highlights the implementation challenges of physical optical platforms [Bontha and Arefin 2024] and Wu et al. show that reliable P1/P4 measurement depends on sufficient image resolution and accurate reflection localization [Wu

Table 2: Eye representations and rendering approaches relevant to modelling Purkinje reflections. Colours rate each entry relative to the other options in its column with respect to one goal, an eye model that can be fitted to a real eye and renders Purkinje reflections for training or fitting a V/A estimator; green is better suited, yellow intermediate, red least suited. Representation: explicit optical surfaces combined with image-fitted neural appearance > explicit surfaces from population parameters that cannot be fitted to an individual (optical eye models, parametric meshes) > purely neural fields without surfaces for reflection or refraction. Reflections: ray-traced or analytic specular reflections > reflection-aware shading (reflected-direction parametrisation, environment maps) > baked view-dependent appearance or rasterised highlights. Input: images needed to personalise the model to a real eye – few images > dense multi-view or light-stage capture > synthetic only, not personalisable. Fidelity: reported agreement with real images – quantitative (PSNR/SSIM) > photorealistic without quantitative comparison > geometric or appearance approximation. Speed: real-time (≥30 fps) > interactive (1–30 fps) > offline. Gaze and Lens: whether eye rotation and lens (accommodative) state are explicit controllable variables.

Ref.	Representation	Reflections	Input	Fidelity	Speed	Gaze	Lens
[?]	Optical eye model	Analytic	Synthetic	Low	Interactive	Yes	Yes
[Wood et al. 2016]	Parametric mesh	Rasterised	Synthetic	Low	Real-time	Yes	No
[Nair et al. 2020]	Parametric mesh	Path-traced	Synthetic	Mid	Offline	Yes	No
[Sharifzadeh 2024]	Optical eye model	Analytic	Synthetic	Low	Interactive	No	No
[Lin et al. 2026]	Optical eye model	Ray-traced	Synthetic	Mid	Offline	Yes	No
[Li et al. 2022]	Hybrid	Analytic	Light stage	High	Offline	Yes	No
[Verbin et al. 2022]	NeRF	Reflection-aware	Multi-view	High	Offline	No	No
[Kerbl et al. 2023a]	3DGS	Baked (SH)	Multi-view	High	Real-time	No	No
[Ruzzi et al. 2023]	NeRF	Baked	Few images	Mid	Offline	Yes	No
[Jiang et al. 2024a]	3DGS	Reflection-aware	Multi-view	High	Real-time	No	No
[Tang et al. 2025]	3DGS	Ray-traced	Multi-view	High	Interactive	No	No
[Wu et al. 2025]	3DGS	Ray-traced	Multi-view	High	Real-time	No	No
[Khademi et al. 2026]	Neural + eye prior	Reflection-aware	Multi-view	High	Offline	Partial	No
Ours	Hybrid	Analytic	Few images	High	Real-time	Yes	Yes

et al. 2023]. These considerations motivate evaluating optical layout, calibration, and sensor-level detectability together.

The present work does not perform end-to-end differentiable optimization. Instead, it uses geometric-optics simulation to compare candidate LED–camera configurations and to predict whether P1–P4 remain visible, sharp, and separable under realizable capture conditions. Differentiable layout optimization and synthetic-to-real adaptation are left as future extensions.

4 Optical Background

4.1 Capturing Purkinje image

Dual Purkinje Image (DPI) tracking can provide higher precision than conventional pupil–corneal-reflection tracking by using the relative motion of P1 and P4 rather than the pupil center [Cornsweet and Crane 1973; Ressmeyer et al. 2026]. However, obtaining reliable Purkinje features from real eye images remains challenging. Wu et al. [Cornsweet and Crane 1973] show that P1 and P4 can be imaged together while P3 is out of focus and P2 largely overlaps with P1; similarly, Ahn et al. [Ahn and Lee 2025] report that P4 is substantially smaller and darker than P1, making its detection less reliable. Therefore, the existence of a valid reflection path does not necessarily guarantee a usable image feature, as its visibility further depends on optical focus, reflection strength, occlusion, and imaging noise [Ahn and Lee 2025; Cornsweet and Crane 1973]. Camera and illumination configuration also affect whether these reflections remain observable across different eye orientations, making Purkinje detectability a joint consequence of ocular geometry and the imaging configuration [Cornsweet and Crane 1973; Lu et al. 2020]. Nevertheless, Purkinje geometry carries useful information about the underlying ocular surfaces, with previous phakometric studies showing that the position and geometry of these reflections can be used to infer ocular surface curvature and alignment [Barry et al. 2001].

4.2 Simulation-to-capture pipeline for near-eye Purkinje imaging

A reliable Purkinje imaging pipeline requires the eye to be modeled as an optical system rather than an appearance object. Since P1–P4 are generated by reflection and refraction at corneal and lenticular surfaces, Arizona/ZEMAX-style schematic eye models are useful because they make accommodation, refractive interfaces, and ray paths physically controllable [Lu et al. 2020][Ozhan et al. 2023]. This also explains why eye-rendering methods often keep the eyeball or cornea as an explicit mesh: specular highlights, corneal refraction, and near-field glints are tied to surface geometry, while NeRF-like fields are more suitable for view synthesis, relighting, and periocular appearance modeling [Khademi et al. 2026; Li et al. 2022; Nair et al. 2020]. Existing Purkinje imaging systems also provide practical guidance for hardware geometry. Both Lu et al. and Ozhan et al. place infrared LEDs close to the camera optical axis to maximize Purkinje visibility while keeping the reflections inside the camera field of view. At the same time, camera positions must balance the visibility of higher-order Purkinje images against eyelid and eyelash occlusions, glint overlap, and sensor saturation. These observations suggest that LED and camera placement is not arbitrary but jointly determines the brightness, separation, and detectability of P1–P4.

Once the capture geometry is determined, synthetic rendering provides an efficient way to evaluate different configurations before physical experiments. RIT-Eyes demonstrates that physically based rendering can generate realistic near-eye images under controllable illumination, while EyeNeRF and subsequent hybrid eye representations further show that explicit corneal geometry is necessary for reproducing specular reflections and refraction, whereas implicit neural representations are better suited for modeling complex periocular appearance and relighting. These observations motivate the use of explicit optical eye models for Purkinje simulation before subsequent NeRF or 3D Gaussian reconstruction, fall outside the field of view, or become unresolvable at the sensor resolution. Thus, the key background principle is that eye geometry, near-field illumination, camera focus/FOV, resolution, and reconstruction representation must be designed together before reliable Purkinje capture and later NeRF/3D Gaussian reconstruction can be expected [Kerbl et al. 2023a]

5 Method

We adopt 3D Gaussian Splatting (3DGS) as the base representation for reconstructing the eye scene [Kerbl et al. 2023a]. The scene is represented by a set of anisotropic 3D Gaussians, each parameterized by its position $\boldsymbol{\mu}$, covariance Σ defined by scale and rotation, opacity α, and spherical harmonics (SH) coefficients for view-dependent appearance [Jiang et al. 2024a; Kerbl et al. 2023a].

During rendering, the 3D Gaussians are projected onto the image plane, depth-sorted, and accumulated using front-to-back α-compositing [Kerbl et al. 2023a]:

$$\mathbf{C}(\mathbf{p}) = \sum_i T_i(\mathbf{p})\alpha_i(\mathbf{p})\mathbf{c}_i(\mathbf{v}),$$

where $\mathbf{c}_i(\mathbf{v})$ denotes the view-dependent color predicted from the SH coefficients. Training jointly optimizes Gaussian positions, scales, rotations, opacities, and SH coefficients using a combination of L_1 and D-SSIM reconstruction losses, while adaptive densification increases the number of Gaussians in regions requiring higher representational capacity [Kerbl et al. 2023a]. We use this standard 3DGS formulation as our baseline and subsequently introduce Purkinje-specific local supervision.

5.1 3D Gaussian Splatting

3D Gaussian Splatting is an explicit 3D scene representation and differentiable rendering method designed for novel view synthesis. The core idea is modeling the geometric structure and viewpoint-dependent appearance of a scene in 3D space using a set of optimizable Gaussian primitives expressed as

$$\mathcal{G}_i = \left\{\boldsymbol{\mu}_i,\ \Sigma_i,\ \alpha_i,\ \mathbf{c}_i\right\}, \tag{1}$$

Here, $\boldsymbol{\mu}_i$ denotes the 3D position of the i-th Gaussian, $\mathbf{s}_i$ its scale, $\mathbf{R}_i$ its rotation, α_i its opacity, and $\mathbf{c}_i$ the spherical harmonics coefficients used to represent view-dependent appearance.

The input to 3DGS consists of a set of multi-view images capturing the same scene from different viewpoints, structure-from-Motion (SfM) estimates the camera parameters and reconstructs a sparse 3D point cloud, which is subsequently used to initialize the Gaussian primitives. [Jiang et al. 2024a; Kerbl et al. 2023a]. The scale and rotation jointly define the covariance matrix as

$$\Sigma_i = \mathbf{R}_i\mathbf{S}_i\mathbf{S}_i^T\mathbf{R}_i^T, \tag{2}$$

where $\mathbf{S}_i$ is a diagonal scaling matrix constructed from $\mathbf{s}_i$, and $\mathbf{R}_i$ determines the orientation of the anisotropic Gaussian.

Once the Gaussian primitives are initialized and parameterized, they are projected onto the image plane, ordered according to depth, and accumulated using front-to-back alpha compositing to generate the rendered image. Standard 3DGS combines a pixel-wise L_1 loss with a D-SSIM loss [Kerbl et al. 2023a]:

$$\mathcal{L}_{\text{GS}} = (1-\lambda)\mathcal{L}_1 + \lambda\mathcal{L}_{\text{D-SSIM}}, \tag{3}$$

where $\mathcal{L}_1$ measures the pixel-wise photometric difference between the rendered and ground-truth images, $\mathcal{L}_{\text{D-SSIM}}$ measures their local structural dissimilarity, and λ controls the relative contribution of the two terms. The reconstruction loss is back-propagated to jointly optimize the Gaussian positions, scales, rotations, opacities, and SH coefficients.

In parallel with parameter optimization, 3DGS performs adaptive density control to dynamically refine the Gaussian population. Gaussians with large positional gradients are cloned when they are sufficiently small and split when they have a large spatial extent, introducing additional primitives in under-reconstructed regions. Conversely, Gaussians with very low opacity or excessively large spatial or screen-space extent are periodically removed. By alternating differentiable rendering, gradient-based optimization, and adaptive density control, the initially sparse Gaussian representation progressively evolves into a dense scene representation capable of high-quality novel-view rendering [Kerbl et al. 2023a]. We learn the appearance of the eye directly from multi-view near-infrared images with 3D Gaussian Splatting (3DGS), without explicit optical modeling, to reproduce the Purkinje reflections and exploit them for accommodation and convergence estimation. The model has two parts: a static canonical 3DGS capturing all view-dependent appearance under a fixed light, including P1, P3 and P4, and a deformation field conditioned on a state vector s = (accommodative demand, horizontal/vertical gaze) capturing eyeball rotation, pupil diameter and the displacement of P3/P4. We rely on the fact that, under a fixed light and moving camera, a curved-surface reflection is multi-view consistent with a virtual image behind the surface, so 3DGS reconstructs each Purkinje reflection as a compact Gaussian cluster whose position becomes a function of s.

Validation proceeds in three stages: a synthetic eye with known cameras; a degradation ladder in which one camera-calibrated degradation (defocus, saturation, compression, exposure drift, pose error, blinks, micro-movements, view density) is added at a time and reconstructed with reflection-preserving variants (Splatfacto, AbsGS) to find where each one breaks; and real captures of one eye under a fixed light with locked focus and exposure, at a grid of known fixation distances and gaze directions, each with a multi-height orbital sweep, with poses solved from all frames and then subsampled for training. We evaluate on held-out views using pupil-region fidelity, P1/P4 centroid error, and the stability of the P1 virtual image at 4 mm behind the corneal apex, not full-image PSNR.

The canonical model is trained on sharp frames and frozen before the deformation field is fitted. At inference, with the camera pose known, we optimize only s per frame to match the rendering to the observed pupil region; the recovered accommodative demand is the accommodation estimate and the difference in binocular gaze angles gives convergence. The resulting (image, s) pairs distill a lightweight encoder for real-time use.

6 Results and Discussions

7 Results and Discussions

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